SHRI RAMSWAROOP MEMORIAL UNIVERSITY



(XCSX606) (INDIAN KNOWLEDGE SYSTEM IN CS)

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The magic of secret codes

‘KATAPAYADI SANKHYA’

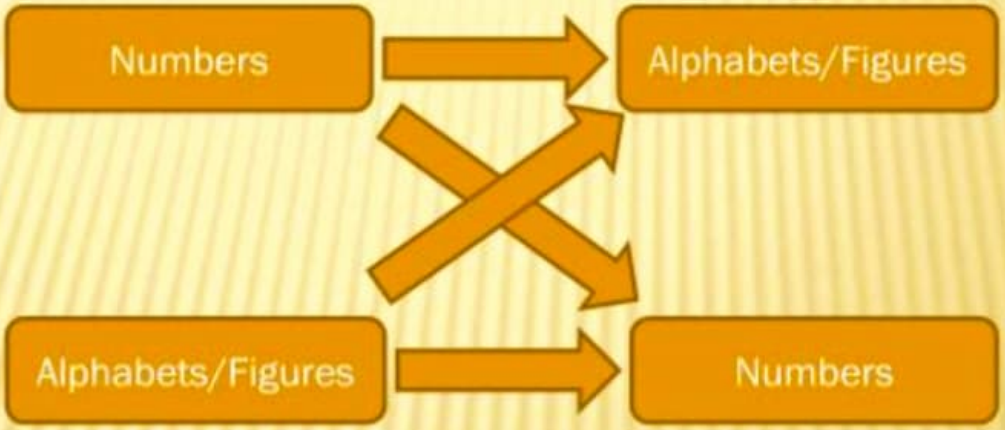
Introduction

The Foundation Course-'Building Mathematical Ability' comprises a section on coding and decoding which describes various methods of encoding and deciphering messages. Pursuant thereto, this presentation is on 'Magic of Secret Codes' that introduces one of the coding methods used in Ancient India.

The presentation focuses on the 'KaTaPaYaDi Sankhya' system of encoding. According to this system, each number is associated with one or more letters such that a sequence of numbers can be encoded in the form of sentences. This system may also be used as memorizing technique akin to the 'peg system'.

Coding

Code is a system of words, alphabets, figures, or other symbols represented by converting into others, especially for secret or secured communication.



As a verb '**Code**' means conversion (of the words of a message) into a particular code in order to convey a secret meaning.

Examples of codes

- **Alphabets to Figures:**
  - Sherlock Holmes and The Story of The Dancing Figures
- **Alphabets to Numbers:**
  - RSA Encoding
- **Numbers to Alphabets:**
  - KaTaPaYaDi Sankhya;
  - Aryabhata System
  - Bhutasamkhya System

KATAPAYADI Sankhya

This is an ancient Indian system of denoting numbers by alphabets. The string of numbers are converted to alphabets such that they form meaningful sentences. The language used in the ancient times was *Sanskrit* and the alphabets were in *devanagri* script.

The ancient coding system followed little-ending system, which is the opposite of what we use today. In other words, the number at unit's place was written on the left followed by the number at 10's place, ending with the highest order on the right.

For example, One Hundred and Thirty Two would be written as '231'.

This system can nonetheless be adapted to the big-ending system that we use today as the only difference in the two systems is in representation. Example 2 discussed later shows this adaptation.

KATAPAYADI Sankhya- Assignment Table

The alphabets are assigned as per the following table:

कटपयादि संख्या - kaTapayAdi for Melakarta Ragam Names & Numbers									
1	2	3	4	5	6	7	8	9	0
क	ख	ग	घ	ङ	च	छ	ज	झ	ञ
ट	ठ	ड	ढ	ण	त	थ	द	ध	न
प	फ	ब	भ	म					
य	र	ल	व	श	ष	स	ह		
ka	kha	ga	gha	nga	cha	Cha	ja	Jha	nya
Ta	Tha	Da	Dha	Na	ta	tha	da	dha	na
pa	pha	ba	bha	ma					
ya	ra	la	va	sha	Sha	sa	ha		

Stand alone vowels denote '0'

Assignment Table (Contd.)

Observations vis-à-vis the Table:

- All consonants without a vowel are ignored;
- Vowels do not change the number allotted. For example, Ka, Ke and Ki all have the value '1';
- More than one alphabet has been allotted to a number;
- '1' has been allotted the maximum alphabets (consonants) and '0' the least;
- Decimal could not be represented by this system

Since any combination of any vowel with a consonant is allotted the same number, this system imparts great flexibility of combining different alphabets to form meaningful sentences and paragraphs. Similarly, allocation of more than one alphabet to a number also gives such flexibility.

KATAPAYADI Sankhya- Example 1

KATAPAYADI Sankhya has been used in ancient times to encode the value of 'pi'

The value of 'pi', up to 17 digits after decimal is:

3.14159265358979324

KATAPAYADI Sankhya- Example

The value of 'pi' can be assigned the following alphabets:

गि gi	प pa	भू bhu	य ya	म ma	धा dha a	र ra	त ta	णि ṇi	ग ga	म ma	ज ja	ध dha	सि sa	धि dhi	बु bu	रा rā	भ bha
3	1	4	1	5	9	2	6	5	3	5	8	9	7	9	3	2	4

In order to make a meaningful sentence, consonants without vowels may also be added.

KATAPAYADI Sankhya- Example 1 (Contd.)

One ancient example used the following encoding:

भ bha	द् d	रा rā	म् m	बु bu	द् d	धि dhi	सि sa	द् d	ध dha	ज ja	न् n	म ma	ग ga	णि ni	त ta	श् sh	र ra	द् d	धा dha	स् s	म ma	य ya	द् d	भू bhu	प pa	गि gi
4	-	2	-	3	-	9	7	-	9	8	-	5	3	5	6	-	2	-	9	-	5	1	-	4	1	3

Since the ancient Indian system used the little-ending way of writing, the alphabets have been allotted to the reverse string of the value of 'pi'.

This gave rise to the following *sloka* for the value of 'pi'

भद्राम्बुद्धिसिद्धजन्मगणितश्रद्धा स्म यद् भूपगी:

(*bhadrāmbuddhisiddhajanmagaṇitaśraddhā sma yad bhūpagīḥ*)

KATAPAYADI Sankhya- Example 1 (Contd.)

Another code for the value of 'pi' as used in ancient India is:

"gopi bhagya madhuvrata
srngiso dadhi sandhiga
khala jivita khatava
gala hala rasandara"

This is a praise for Lord Krishna and translates into:

"O Lord anointed with the yoghurt of the milkmaids' worship (Krishna), O savior of the fallen, O master of Shiva, please protect me."

This encoding gives the value of 'pi' up to 31 places after decimal i.e. **3.14159265358979323846264338327 92**

KATAPAYADI Sankhya

KaTaPaYaDi Sankhya can also be used to memorize long numbers such as phone numbers, account numbers etc. and for writing down in a secured fashion.

The memorizing technique is akin to the 'peg-system' of memorizing in which the numbers are 'pegged' to a picture or a story for easy recall.

THANK YOU